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Senior Embedded Software Engineer

Speciality :

**real time
software architecture**

[LinkedIn](#)



>> [Github](#) <<

↘ Academic Background

2024-2025 : AI Deep Learning certifications : Convolutional Neuronal Networks ; Structuring Machine Learning Projects ; Neuronal Networks and Deep Learning design by Andrew Ng

2007-2011 : Engineer in ENSEA. Ecole Nationale Supérieure de l'Électronique et de ses Applications

↘ Experience summary

Embedded Software Architect and Technical Lead with 14+ years of experience designing safety-critical embedded systems across aerospace, defense, medical and industrial sectors. Strong expertise in C/C++, Rust, Linux, Yocto, RTOS, functional safety, hypervisors and heterogeneous SoCs. Proven track record leading architecture decisions, optimizing system performance and mentoring engineering teams.

My motto : "I step back to understand the broader context and evaluate trade-offs before committing to a solution."

↘ Software & electronics Skills

► Core Expertise : Embedded Software Architecture, Functional Safety, RTOS, Linux BSP, Hypervisors, Heterogeneous SoCs, Performance Optimization, Hardware/Software Integration.

► Programming Languages : C (C89, C99), C++ (11-20) ; Rust ; Assembly ARM ; Java (7+) ; Python (Jinja2, Django, Pyro) ;
I have a preference for strongly typed languages.

► Operating system : Embedded Linux (Yocto), Linux PREEMPT_RT, FreeRTOS, SafeRTOS, Autosar, PikeOS (ARINC 653), VxWorks, Android (AOSP).

Strong expertise in the Linux boot chain (U-Boot, Device Tree, kernel configuration and BSP integration) ;

Experience with Type-1 hypervisors (Bao), heterogeneous SoCs and hardware partitioning.

► Tools & Frameworks :

Build Systems : Makefile, Maven, Gradle, CMake and Conan ;

Graphics & Multimedia : OpenCv, Cuda, OpenGL, FFmpeg, Gstreamer ;

Development Tools : Jenkins (Pipeline script & declarative), Docker, QEMU ;

Version Control : svn, git, IBM RTC ;

Testing & Mocking : Mockito, Junit (java), Google Test, IBM RTRT, Parasoft (C/C++), PyTest (Python) ;

Publish/Subscribe middleware expertise : DDS (Rti & Prismtech) ; ZMQ ;

► Electronics Experience :

Platforms : MSP430 ; ARM : imx6 Cortex-A9, Microsemi SmartFusion2 Cortex-M3 ; NXP S32G2 (Cortex-A53 & M7) ;

Driver development : CAN, UART, SPI, I2C, GPIO, Ethernet, MIL-STD-1553, etc... ;

Electronics : small PCB design, noise treatment, schematics analysis, etc ;

Debugging : GDB, GDB/MI, Trace32.

↘ Foreign languages

► French : Native

► English: Advanced (TOEIC : 785pts)

► Japanese : Beginner (JLPT N5 ~ N4)

↘ Work experience

Company **PowerX / パワーエックス** – Tokyo, Japan – october 2025 – today

Product Owner / Technical Team Lead – Industrial PLC Platform (SafeRTOS, Yocto RT, Type-1 Hypervisor)

Designed a safety-certified PLC platform for industrial applications based on the heterogeneous NXP S32G2 SoC.

Defined the system architecture and key design decisions based on functional safety and certification requirements.

Migration from FreeRTOS to SafeRTOS.

Designed the secure and safety-compliant boot sequence.

Designed the integration of the Bao Type-1 static partitioning hypervisor to run multiple operating systems on Cortex-A cores.

Benchmarked the Linux RT scheduler by configuring CPU affinity, task periods, priorities, watchdog timeouts and workload.

Designed hardware partitioning using the NXP XRDC module by assigning peripherals and memory resources to dedicated domains.

Developed Lauterbach Trace32 automation scripts for development, debugging and validation.

Developed an AI-assisted Hardware-in-the-Loop (HIL) validation bench to automate testing of advanced hardware features.

Developed a deep understanding of the AUTOSAR architecture and its mechanisms to implement an AUTOSAR-like layer for our codebase. Defined peripheral, domain, and hardware configurations to generate boilerplate code between modules automatically.

Technologies : C, Rust, Yocto Linux, SafeRTOS, FreeRTOS, Autosar, Bao Hypervisor, U-Boot, Docker, Trace32, NXP S32G2, XRDC.

Human aspect : Senior member of the team, introduced industrial software development processes . Technical lead for U-boot, Linux and Hypervisor architecture.

Designed a generic binary runtime based on Rust and WASI technologies

Selected the most suitable WASI runtime based on safety, implementation language, performance, maintainability and project maturity.

Researched and benchmarked multiple WASM runtimes supporting different WASI interfaces and WebAssembly binaries produced from C, C++, Rust, Python, and other languages.

Designed and evaluated custom host interfaces to expose platform-specific features (e.g., networking, UI, ICMP ping).

Integrated the selected runtime into a Yocto-based Linux distribution.

Presented and successfully advocated for the solution, leading to its adoption by engineering management.

Company **Rokken / ロッケン** – Osaka, Japan – march 2024 – september 2025 – 1½ Years

ISO 13485 (medical software japanese certification) – Developer (C++/Cuda/TensorRT) for Medical Software

Development of video application for medical operation using AI to overlay operation risk and organ, configuration of Linux image to package it and creation of a light usb installer. Video source : live streaming and video playback.

AI model integration, shader development, Cuda pipeline design and their optimization for video and AI overlay streaming.

Key achievements : Move video frame manipulation from CPU to GPU. Reduce latency from 85ms to less than 60ms. Reduce frame processing by 20% (use of Nvidia Nsight System). Increase FPS from 55 to 75. Reduce code size by 30%.

Technologies : C++ 17, OpenGL, GLSL, Cuda & Qt.

OS and Build system : Ubuntu, Debian, Docker, Vcpkg and Cmake.

Human aspect : Agile methodology, mentoring junior developers and providing customer guidance.

Product Owner (Rust / Yocto / Kernel) for Medical Urometer

Integrated a Tinker Board 3S SBC with a custom hardware daughterboard. Electronics daughter board integration from customer company. Linux kernel configuration, device tree modification, usage of GPIO, I2C, PWM, MIPI-DSI and Ethernet. Yocto configuration (layers, build configuration, package version update). HMI technology based on Rust Iced library with hardware acceleration. Fix of Rust GLES wrapper library.

Key achievements : learn Yocto from scratch in a week. Integrate last Rust version. Patch Linux DTS for hardware integration. Integrate most famous Rust GUI (Iced) and its hardware acceleration by patching Rust wgpu crate library. Software architecture.

OS and Build system : Yocto Linux.

Company **AIRBUS Helicopters** – Marignane – june 2022 – february 2024 – 2 Years

DO178 (avionics certification) – Embedded Software engineer (C/Rust) for RTOS

Developed software in C, Rust and ARM assembly (parallel computation SIMD) across multiple RTOS partitions. Integration of the supplier's middleware. Continuous integration setup (push → build → flash → test). Use of Airbus proprietary tools and successfully implemented and open-source build pipeline as a proof of concept with the couple Conan / Cmake. Setup of QEMU for running unit tests. Configuration & design proposition. Tool maintenance and creation of Python api layer for existing tool with Pyro python library in order to automatize test. Mentored a junior engineer.

Key achievements : Reduce time consumption for CRC calculation by 83% using SIMD instruction. Complete proof of concept for the usage of Conan build tool as a potential new tool instead Airbus proprietary tool. Reduce development workflow time by 95% (dev + compile + flash and test). Unblocked the German team in the integration of a partition developed in Rust, they had been blocked for 3 months and resolved the issue in a day.

Technologies : OS : PikeOs with layer Apex Arinc653. Real time, Cmake, Conan, Qemu, Python. Agile methodology & Scrum meeting.

Company **MBDA** – Bourges – december 2019 – june 2022 – 2½ years

Software manager / Embedded Software Engineer (C) for RTOS

Development of low level software for the Spectra Program. A complete counter-measure system for air dominance. System based on a multi-product architecture operating with ARMv7 processors (A9 and M3). Driver for bare metal and real time OS environment (1553, Eeprom, Emmc, spi, Ethernet, etc...) with highly configurable delivery. Driver integration and work with electronics team to proceed to problem-solving. Team leader and management of software developer.

Took over an existing project in parallel to complete it due to a colleague departure.

Processor : imx6 (ARMv7 A9); RTSOS: PikeOS. Fix bug, add unitary test & integration test. Weekly meeting, regular delivery and apply the documentation process.

Key achievements : Improvement of tools to make them suitable for industrialization. Reduce the validation time of electronics cards from 2 weeks to 2 hours. Script creation for proprietary tools to integrate them into a build process. Work on code sharing : remove of all copy/paste code by using of all of the feature of our source code management (on my second project, the volume of project code decrease by 90%).

Technologies : Other tools : Cpptest, IBM Rational Test RealTime. Source code management tool : IBM RTC. MISRA 2012 development rules.

Company **Parrot Faurecia Automotive** – Paris – september 2017 – december 2019 – 2 years

Embedded Technical Leader for bench software, Lead dev for C/C++, Support Java/Python/Android

Main software : responsible of the power management system. Development of C++/Java service in Android (AOSP). Travel to the McLaren HeadQuarter in England for integration purpose. Setup of the continuous integration by creation of Jenkins job.

Bench software : technical lead and management of one developer. Configure a Linux kernel and prepare an embedded Linux image to fully test the electronics board to ensure its proper functioning. Read electronics schematics and documentation to drive development.

Side project : creation of a C++ generator in Java to generate client side service for IPC message exchange & expose this message to Qt QML file in order to develop HMI more efficiently.

Tools development : Development of a Python package and a VsCode plugin to generate VsCode config file to load/unload Android target in the IDE in a fast way.

Key achievements : manage small team, backport AIDL mechanism from Android 8 to Android 5. Save time for development team by the creation of a C++ generator code.

Miscellaneous : operating system hypervisor notion, agile methodology & Scrum meeting.

Safran Electronics and Defense (Safran Engineering Services) – Massy – feb. 2014 – sept. 2017 – 2½ years

(2016 – 2017) **Lead developer** on C++/Java, Gstreamer/OpenGL/DDS & Android support

Development of a cross-platform Vetronics (Windows, Linux & Android) for military vehicle.

The main goal was to simulate multiple vehicles on a battlefield. The vehicle can be either tank or reconnaissance vehicle and its equipment was virtualized. Like canon, camera and sensor.

Based on publish/subscribe messaging middleware, service has been developed and interacts each other. This service can be video streaming, interactor driver (like gaming controller, wheel drive, etc...), command control, etc...

(2014 – 2016) C++/Java developer

Cross-platform software (Windows, Linux, Android) for military vehicle as proof of concept for next project. Two times, prepare a demonstration for the Eurosatory exhibition event, a professional forum to show military product.

Key achievements : Project was in partnership with Thales and their team start to use DDS. I provide a huge help and advise on how debug DDS and how define a good data model. A lot of code was reused from the previous project. Debugged the Thales platform test, resolving an issue they had faced for three months.

Technologies : Publish-Subscribe DDS RTI, component oriented programming, Serial/Can/Usb/Ethernet/Wifi connectics, Gstreamer, OpenCv, OpenGL, Jni/Jna, Maven, PID control, Augmented reality, Kernel Linux & modules/driver

Safran Aircraft Engines (Safran Engineering Services) – Villaroche – feb. 2012 – feb. 2014 – 2 years

Industrial software developer (Java / .Net / C++)

Developed Java desktop applications for aircraft manufacturing resource management and engine health monitoring, using Swing, JavaFX, Oracle, and UDP communication. Also developed a WPF (.NET) demonstrator for trade exhibitions.

Intervener at **ENSEA** for Java practical work – Cergy – 2014 – 2 years (1 month each year, 4h by week)

Intervener at engineering school

Portfolio

- Bicycle tachometer (2004);
- Magic the gathering website to searching cards, building decks, and play online (2006);
- Android application for my engineering school (well known among students and widely used) (2007 - 2013);
- Windows tool to help color-blind people identify the color they use (2019);
- Fix a bug in Cmake and push the patch to the community (2021);
- Development of feature for Conan build system (2022);
- Small Rust + Vulkan program to discover Voxel modeling (2025);
- Python application for reviewing Japanese vocabulary (active) (2025 ~ nowadays);
- Hardware and modeling project for rocket launches (active) (2026 ~ nowadays).

Hobbies

- Climbing, swimming help me of keeping focus and maintain my physical endurance;
- Magic : the Gathering competitive player for its strategic and analytical gameplay to solve problem and manage resources. Also a game I played since my college time and become a real passion;
- Development for personal project to keep me update of new technologies. Like AI image processing, things like that...
- Reading old books and learning Japanese cultivating my cultural awareness.